tinyriscv_project_mod 严格验证与代码规范审查报告

生成时间：2026-05-02 19:46:40
工程路径：E:\project-vlsi-20260422\tinyriscv_project_mod
指导书：c:\Users\Lenovo\Desktop\数字大规模集成电路-project.pdf
验证方式：Icarus Verilog 编译/仿真 + 静态源码扫描 + 人工芯片工程规范审查

1. 总结结论

严格结论：当前 RTL 功能仿真通过，但还不能认定为流片签核完成。

本次从头到尾重新运行了 filelist 检查、全部 RTL 编译、SoC 顶层 elaboration、七个 testbench、静态敏感语法扫描、源码行数统计和 module/endmodule 配平。所有可运行仿真均为 PASS，命令退出码均为 0。

从芯片工程角度，仍有若干必须在 tapeout 前关闭的事项：综合/STA/CDC/DFT/APR/DRC/LVS/GDS 尚未提供，pad-level 多芯片 IO Ring wrapper 尚未实现，JTAG CDC、EX 关键路径、默认学号占位、warning-free lint 仍需处理。

2. 指导书要求逐项对照

编号	指导书要求	当前严格状态	证据与说明
R1	每人独立对 tinyriscV 进行修改，添加课程要求的拓展指令。	PARTIAL PASS	RTL 已实现 sID/rT/if 三条工程声明的自定义指令，并通过 tb_custom_unit；但指导书未给出拓展指令编码原表，真实课程编码仍需和助教确认。
R2	在 FPGA 上进行功能验证（助教提供运行指令）。	NOT CLOSED	仓库内已通过 FPGA 侧 bridge 模型仿真，但未进行真实 FPGA 板级运行和助教脚本验证。
R3	进行综合、仿真、后端等工作，提供 DRC 与 LVS clean 的 GDS 文件。	NOT CLOSED	仿真已通过；本环境无 Verilator/Yosys，且仓库无 PDK、综合脚本、APR、DRC/LVS 或 GDS。
R4	对流片回来的芯片进行功能与性能测试。	FUTURE	按指导书时间，回片测试在 2026.11-2026.12；当前无法完成。
R5	流片前两颗及更多芯片放在一个 IO Ring 内部共用 IO，通过 chip_sel 选择工作的芯片。	PARTIAL PASS	当前 SoC 有 chip_sel_i，并通过 tb_chip_sel；但 pad-level 多芯片 IO Ring wrapper 和输出 OE/mux 未在仓库中。

3. 运行结果矩阵

步骤	检查内容	状态	退出码
01_filelist_check	filelist 完整性检查	PASS	0
02_all_rtl_compile	全部 RTL 源码编译	PASS	0
03_top_elab	SoC 顶层 elaboration	PASS	0
04_tb_custom_unit	自定义指令单元 sID/rT/if	PASS	0
05_tb_pwm	PWM 外设	PASS	0
06_tb_i2c_lm75	I2C/LM75 外设	PASS	0
07_tb_mem_bridge	芯片侧/FPGA 侧片外存储桥	PASS	0
08_tb_soc_load	SoC 片外取指与 load	PASS	0

步骤	检查内容	状态	退出码
09_tb_soc_mem_ops	SoC 片外 load/store/byte/halfword	PASS	0
10_tb_chip_sel	chip_sel 选择逻辑	PASS	0
11_static_disallowed_constructs	禁用/敏感语法扫描	CHECKED	0
12_static_timing_directives	timescale/default_nettype 扫描	CHECKED	0
13_static_todos_placeholders	TODO/占位项扫描	CHECKED	0
14_static_tristate_inout	inout/三态扫描	CHECKED	0
15_line_counts	源码行数统计	PASS	0
16_module_balance	module/endmodule 配平	PASS	0

说明：静态扫描步骤的 CHECKED 表示扫描命令执行成功且命中项已在第 5 节解释，不等价于无风险。

4. 芯片工程师规范审查结论

严重级别	问题	工程判断
HIGH	尚未达到流片签核闭环	缺综合、STA、CDC/RDC、DFT、APR、DRC/LVS、GDS 和 pad-level IO Ring；因此当前不是 tapeout-ready。
HIGH	多芯片共 IO Ring 只完成 RTL 选择，未完成 pad 互斥	chip_sel_i 能让未选芯片复位，但共享 pad 时还需要输出 mux/OE，避免多个 die 同时驱动同一 IO。
MEDIUM	EX 组合路径偏大	core/ex.v 大 case + 组合乘法可能限制频率；高频目标建议多周期乘法或流水化。
MEDIUM	JTAG_TCK CDC 需签核	debug/jtag_driver.v 使用 jtag_TCK，core 用 clk；虽有 full_handshake，仍需 CDC 工具确认。
MEDIUM	RTL 无统一 `default_nettype none`，多数 RTL 无显式 timescale	这不是综合功能错误，但不符合严格流片 lint 风格。
LOW	Icarus warning 未完全清零	regs/pwm 数组组合读触发 Icarus sensitivity warning；功能仿真 PASS，但建议清理到 warning-free。
LOW	占位学号未替换	core/custom_unit.v DEFAULT_ID_* 目前是示例值。

5. 逐文件代码规范审查

文件	结论	审查意见
core/defines.v	PASS	集中定义 reset、地址空间、RV32I/M/CSR、自定义指令宏。建议后续加入 default_nettype none。
core/pc_reg.v	PASS	PC reset/hold/jump 逻辑清晰，组合路径短。
core/if_id.v	PASS	流水寄存器用 gen_pipe_dff，hold 行为统一。
core/id.v	PASS	译码组合块有默认赋值，未见锁存器风险；自定义指令译码已覆盖。
core/id_ex.v	PASS	流水寄存器结构规则。
core/ex.v	PASS WITH FREQ RISK	功能仿真通过；但 EX 是大组合块，并含组合乘法 mul_temp=mul_op1*mul_op2，是主要频率风险。

文件	结论	审查意见
core/regs.v	PASS WITH WARNING	功能可用；Icarus 对 regs 数组组合读给 warning。GPR 未 reset，ASIC 可接受但 boot 程序要避免读未初始化寄存器。
core/csr_reg.v	PASS / NEEDS MORE TEST	编译通过，结构完整；当前回归未专门覆盖 CSR 指令和异常路径。
core/clint.v	PASS / NEEDS MORE TEST	编译通过；当前回归未专门覆盖异常、mret、中断时序。
core/div.v	PASS / NEEDS MORE TEST	编译通过；当前回归未专门覆盖 DIV/REM 边界条件。
core/custom_unit.v	PASS WITH SIGNOFF ITEM	sID/rT/if 单测通过；DEFAULT_ID_* 仍是占位学号，流片前必须替换。
core/ctrl.v	PASS	RIB full hold 与 PC-only hold 优先级清楚。
core/rib.v	PASS	片外事务 FSM 锁 owner，解决多周期 bridge 与取指/访存竞争；建议后续加 formal liveness。
core/tinyriscv.v	PASS	片外 load 延迟写回修复有效；建议扩大 RV32I/M 指令级回归。
debug/jtag_driver.v	COMPILE PASS / CDC REVIEW REQUIRED	JTAG_TCK 时钟域存在，使用 full_handshake；流片前必须跑 CDC 约束和检查。
debug/jtag_dm.v	COMPILE PASS / NEEDS TEST	DM 编译通过；当前回归未跑 JTAG transaction test。
debug/jtag_top.v	COMPILE PASS	JTAG driver 和 DM 集成。
perips/uart_shared.v	PASS / NEEDS BFM	MMIO/custom TX 共享路径结构清楚；当前未做真实 UART bit-level BFM 全覆盖。
perips/uart_debug.v	COMPILE PASS / NEEDS SCRIPT MATCH	下载 FSM 编译通过；需与助教 UART 下载脚本逐字节对齐。
perips/uart.v	COMPILE PASS / LEGACY	原 UART 模块未在当前 SoC 顶层实例化，可保留参考或后续移除。
perips/pwm.v	PASS WITH WARNING	PWM 单测通过；Icarus 对数组读 warning，可接受但 lint 时建议清理。
perips/i2c_lm75.v	PASS / PROTOCOL LIMITED	I2C 单测通过；SDA 三态只在 pad-facing inout；ACK/NACK 错误处理较简化。
perips/chip_mem_bridge.v	PASS	固定 8-bit 帧协议测试通过；无 valid/ready/CRC，板级需关注鲁棒性。
soc/tinyriscv_soc_top.v	PASS / IO RING NOT CLOSED	顶层集成和 chip_sel 仿真通过；pad-level 多芯片 IO Ring mux/OE 未实现。
utils/full_handshake_tx.v	COMPILE PASS / CDC REVIEW	握手发送端结构正常；需 CDC signoff。
utils/full_handshake_rx.v	COMPILE PASS / CDC REVIEW	握手接收端结构正常；需 CDC signoff。
utils/gen_buf.v	PASS	参数化同步 buffer。
utils/gen_dff.v	PASS	参数化 DFF 模块，module/endmodule 数量匹配。
fpga/fpga_mem_bridge.v	FPGA/SIM PASS	作为 FPGA 侧模型含 initial/$readmemh；不应作为 ASIC chip RTL 直接综合。

6. 静态扫描解释

禁用/敏感语法扫描命中项需要区分芯片 RTL 与 FPGA/仿真模型：

- **fpga/fpga_mem_bridge.v:39,43** 使用 initial/$readmemh：这是 FPGA 侧 ROM/RAM 模型，不应作为 ASIC chip RTL 直接综合。
- **perips/i2c_lm75.v:67** 使用 1'bz：该三态位于 I2C SDA pad-facing inout，符合开漏 I2C 语义；ASIC 后端需映射到开漏/三态 pad cell。
- **core/rib.v:9** 的 WAIT 只是注释命中，不是 Verilog wait 语句。
- RTL 目录未发现 casex/casez/force/release/$display/$finish/$stop/1'bx 等高风险构造。
- 只有 sim testbench 有 `timescale`，RTL 没有统一 timeunit/default_nettype；建议后续补充项目级规范。

7. 运行日志原文

以下为本次运行结果的原始日志。较长日志在本 PDF 中保留开头和结尾；完整日志同时保存在 `doc/strict_verification_20260502/logs/`。

01_filelist_check - filelist 完整性检查

```
===== COMMAND =====
internal Python filelist check
===== OUTPUT =====
+incdir+./core -> compiler option
+incdir+./utils -> compiler option
+incdir+./debug -> compiler option
+incdir+./perips -> compiler option
+incdir+./soc -> compiler option
./utils/gen_dff.v -> True
./utils/gen_buf.v -> True
./utils/full_handshake_rx.v -> True
./utils/full_handshake_tx.v -> True
./core/clint.v -> True
./core/csr_reg.v -> True
./core/ctrl.v -> True
./core/div.v -> True
./core/regs.v -> True
./core/pc_reg.v -> True
./core/if_id.v -> True
./core/id.v -> True
./core/id_ex.v -> True
./core/ex.v -> True
./core/custom_unit.v -> True
./core/rib.v -> True
./core/tinyriscv.v -> True
./debug/jtag_driver.v -> True
./debug/jtag_dm.v -> True
./debug/jtag_top.v -> True
./perips/uart_shared.v -> True
./perips/uart_debug.v -> True
./perips/chip_mem_bridge.v -> True
./perips/pwm.v -> True
./perips/i2c_lm75.v -> True
./soc/tinyriscv_soc_top.v -> True
===== EXIT_CODE: 0 =====
```

02_all_rtl_compile - 全部 RTL 源码编译

```
===== COMMAND =====
iverilog -g2012 -Wall -I core -o all_rtl_compile.vvp core\clint.v core\csr_reg.v core\ctrl.v core\custom_unit.v core\defines.v core\div.v
core\ex.v core\id.v core\id_ex.v core\if_id.v core\pc_reg.v core\regs.v core\rib.v core\tinyriscv.v debug\jtag_dm.v debug\jtag_driver.v
debug\jtag_top.v perips\chip_mem_bridge.v perips\i2c_lm75.v perips\pwm.v perips\uart.v perips\uart_debug.v perips\uart_shared.v
soc\tinyriscv_soc_top.v utils\full_handshake_rx.v utils\full_handshake_tx.v utils\gen_buf.v utils\gen_dff.v fpga\fpga_mem_bridge.v
===== OUTPUT =====
core\regs.v:74: warning: @* is sensitive to all 32 words in array 'regs'.
core\regs.v:86: warning: @* is sensitive to all 32 words in array 'regs'.
core\regs.v:95: warning: @* is sensitive to all 32 words in array 'regs'.
perips\pwm.v:75: warning: @* is sensitive to all 4 words in array 'period'.
perips\pwm.v:76: warning: @* is sensitive to all 4 words in array 'period'.
perips\pwm.v:77: warning: @* is sensitive to all 4 words in array 'period'.
perips\pwm.v:78: warning: @* is sensitive to all 4 words in array 'period'.
perips\pwm.v:79: warning: @* is sensitive to all 4 words in array 'high_time'.
perips\pwm.v:80: warning: @* is sensitive to all 4 words in array 'high_time'.
perips\pwm.v:81: warning: @* is sensitive to all 4 words in array 'high_time'.
perips\pwm.v:82: warning: @* is sensitive to all 4 words in array 'high_time'.
===== EXIT_CODE: 0 =====
```

03_top_elab - SoC 顶层 elaboration

```
===== COMMAND =====
iverilog -g2012 -Wall -o sim_project.vvp -s tinyriscv_soc_top -f filelist_project.f
===== OUTPUT =====
./core/regs.v:74: warning: @* is sensitive to all 32 words in array 'regs'.
./core/regs.v:86: warning: @* is sensitive to all 32 words in array 'regs'.
./core/regs.v:95: warning: @* is sensitive to all 32 words in array 'regs'.
./perips/pwm.v:75: warning: @* is sensitive to all 4 words in array 'period'.
./perips/pwm.v:76: warning: @* is sensitive to all 4 words in array 'period'.
./perips/pwm.v:77: warning: @* is sensitive to all 4 words in array 'period'.
./perips/pwm.v:78: warning: @* is sensitive to all 4 words in array 'period'.
./perips/pwm.v:79: warning: @* is sensitive to all 4 words in array 'high_time'.
./perips/pwm.v:80: warning: @* is sensitive to all 4 words in array 'high_time'.
./perips/pwm.v:81: warning: @* is sensitive to all 4 words in array 'high_time'.
./perips/pwm.v:82: warning: @* is sensitive to all 4 words in array 'high_time'.
===== EXIT_CODE: 0 =====
```

04_tb_custom_unit - 自定义指令单元 sID/rT/if

```
===== COMMAND =====
iverilog -g2012 -Wall -I core -o sim_custom_unit.vvp sim\tb_custom_unit.v core\custom_unit.v && vvp sim_custom_unit.vvp
===== OUTPUT =====
core\custom_unit.v:12: warning: timescale for custom_unit inherited from another file.
sim\tb_custom_unit.v:1: ...: The inherited timescale is here.
PASS: tb_custom_unit
sim\tb_custom_unit.v:110: $finish called at 435000 (1ps)
===== EXIT_CODE: 0 =====
```

05_tb_pwm - PWM 外设

```
===== COMMAND =====
iverilog -g2012 -Wall -I core -o sim_pwm.vvp sim\tb_pwm.v perips\pwm.v && vvp sim_pwm.vvp
===== OUTPUT =====
perips\pwm.v:16: warning: timescale for pwm inherited from another file.
sim\tb_pwm.v:1: ...: The inherited timescale is here.
perips\pwm.v:75: warning: @* is sensitive to all 4 words in array 'period'.
perips\pwm.v:76: warning: @* is sensitive to all 4 words in array 'period'.
perips\pwm.v:77: warning: @* is sensitive to all 4 words in array 'period'.
perips\pwm.v:78: warning: @* is sensitive to all 4 words in array 'period'.
perips\pwm.v:79: warning: @* is sensitive to all 4 words in array 'high_time'.
perips\pwm.v:80: warning: @* is sensitive to all 4 words in array 'high_time'.
perips\pwm.v:81: warning: @* is sensitive to all 4 words in array 'high_time'.
perips\pwm.v:82: warning: @* is sensitive to all 4 words in array 'high_time'.
PASS: tb_pwm
sim\tb_pwm.v:68: $finish called at 116000 (1ps)
===== EXIT_CODE: 0 =====
```

06_tb_i2c_lm75 - I2C/LM75 外设

```
===== COMMAND =====
iverilog -g2012 -Wall -I core -o sim_i2c.vvp sim\tb_i2c_lm75.v perips\i2c_lm75.v && vvp sim_i2c.vvp
===== OUTPUT =====
perips\i2c_lm75.v:14: warning: timescale for i2c_lm75 inherited from another file.
sim\tb_i2c_lm75.v:1: ...: The inherited timescale is here.
PASS: tb_i2c_lm75
sim\tb_i2c_lm75.v:58: $finish called at 871000 (1ps)
===== EXIT_CODE: 0 =====
```

07_tb_mem_bridge - 芯片侧/FPGA 侧片外存储桥

```
===== COMMAND =====
iverilog -g2012 -Wall -I core -o sim_mem_bridge.vvp sim\tb_mem_bridge.v perips\chip_mem_bridge.v fpga\fpga_mem_bridge.v && vvp
sim_mem_bridge.vvp
===== OUTPUT =====
perips\chip_mem_bridge.v:24: warning: timescale for chip_mem_bridge inherited from another file.
sim\tb_mem_bridge.v:1: ...: The inherited timescale is here.
fpga\fpga_mem_bridge.v:6: warning: timescale for fpga_mem_bridge inherited from another file.
sim\tb_mem_bridge.v:1: ...: The inherited timescale is here.
PASS: tb_mem_bridge
sim\tb_mem_bridge.v:84: $finish called at 540000 (1ps)
===== EXIT_CODE: 0 =====
```

08_tb_soc_load - SoC 片外取指与 load

```
===== COMMAND =====
iverilog -g2012 -Wall -I core -o sim_soc_load.vvp -f filelist_project.f sim\tb_soc_load.v fpga\fpga_mem_bridge.v && vvp sim_soc_load.vvp
```

```
===== OUTPUT =====
fpga\fpga_mem_bridge.v:6: warning: timescale for fpga_mem_bridge inherited from another file.
sim\tb_soc_load.v:1: ...: The inherited timescale is here.
warning: Some design elements have no explicit time unit and/or
       : time precision. This may cause confusing timing results.
       : Affected design elements are:
       :   -- module chip_mem_bridge declared here: ./perips/chip_mem_bridge.v:24
       :   -- module clint declared here: ./core/clint.v:22
       :   -- module csr_reg declared here: ./core/csr_reg.v:20
       :   -- module ctrl declared here: ./core/ctrl.v:21
       :   -- module custom_unit declared here: ./core/custom_unit.v:12
       :   -- module div declared here: ./core/div.v:22
       :   -- module ex declared here: ./core/ex.v:21
       :   -- module full_handshake_rx declared here: ./utils/full_handshake_rx.v:23
       :   -- module full_handshake_tx declared here: ./utils/full_handshake_tx.v:23
       :   -- module gen_en_dff declared here: ./utils/gen_dff.v:125
       :   -- module gen_pipe_dff declared here: ./utils/gen_dff.v:18
       :   -- module gen_rst_0_dff declared here: ./utils/gen_dff.v:46
       :   -- module gen_rst_1_dff declared here: ./utils/gen_dff.v:72
       :   -- module gen_rst_def_dff declared here: ./utils/gen_dff.v:98
       :   -- module gen_ticks_sync declared here: ./utils/gen_buf.v:18
       :   -- module i2c_lm75 declared here: ./perips/i2c_lm75.v:14
       :   -- module id declared here: ./core/id.v:21
       :   -- module id_ex declared here: ./core/id_ex.v:20
       :   -- module if_id declared here: ./core/if_id.v:20
       :   -- module jtag_dm declared here: ./debug/jtag_dm.v:27
       :   -- module jtag_driver declared here: ./debug/jtag_driver.v:23
       :   -- module jtag_top declared here: ./debug/jtag_top.v:20
       :   -- module pc_reg declared here: ./core/pc_reg.v:20
       :   -- module pwm declared here: ./perips/pwm.v:16
       :   -- module regs declared here: ./core/regs.v:20
       :   -- module rib declared here: ./core/rib.v:16
       :   -- module tinyriscv declared here: ./core/tinyriscv.v:20
       :   -- module tinyriscv_soc_top declared here: ./soc/tinyriscv_soc_top.v:13
       :   -- module uart_debug declared here: ./perips/uart_debug.v:31
       :   -- module uart_shared declared here: ./perips/uart_shared.v:12
./core/regs.v:74: warning: @* is sensitive to all 32 words in array 'regs'.
./core/regs.v:86: warning: @* is sensitive to all 32 words in array 'regs'.
./core/regs.v:95: warning: @* is sensitive to all 32 words in array 'regs'.
./perips/pwm.v:75: warning: @* is sensitive to all 4 words in array 'period'.
./perips/pwm.v:76: warning: @* is sensitive to all 4 words in array 'period'.
./perips/pwm.v:77: warning: @* is sensitive to all 4 words in array 'period'.
./perips/pwm.v:78: warning: @* is sensitive to all 4 words in array 'period'.
./perips/pwm.v:79: warning: @* is sensitive to all 4 words in array 'high_time'.
./perips/pwm.v:80: warning: @* is sensitive to all 4 words in array 'high_time'.
./perips/pwm.v:81: warning: @* is sensitive to all 4 words in array 'high_time'.
./perips/pwm.v:82: warning: @* is sensitive to all 4 words in array 'high_time'.
PASS: tb_soc_load
sim\tb_soc_load.v:105: $finish called at 3050000 (1ps)
===== EXIT_CODE: 0 =====
```

09_tb_soc_mem_ops - SoC 片外 load/store/byte/halfword

```
===== COMMAND =====
iverilog -g2012 -Wall -I core -o sim_soc_mem_ops.vvp -f filelist_project.f sim\tb_soc_mem_ops.v fpga\fpga_mem_bridge.v && vvp
sim_soc_mem_ops.vvp
===== OUTPUT =====
fpga\fpga_mem_bridge.v:6: warning: timescale for fpga_mem_bridge inherited from another file.
sim\tb_soc_mem_ops.v:1: ...: The inherited timescale is here.
warning: Some design elements have no explicit time unit and/or
       : time precision. This may cause confusing timing results.
       : Affected design elements are:
       :   -- module chip_mem_bridge declared here: ./perips/chip_mem_bridge.v:24
       :   -- module clint declared here: ./core/clint.v:22
       :   -- module csr_reg declared here: ./core/csr_reg.v:20
       :   -- module ctrl declared here: ./core/ctrl.v:21
       :   -- module custom_unit declared here: ./core/custom_unit.v:12
       :   -- module div declared here: ./core/div.v:22
       :   -- module ex declared here: ./core/ex.v:21
       :   -- module full_handshake_rx declared here: ./utils/full_handshake_rx.v:23
       :   -- module full_handshake_tx declared here: ./utils/full_handshake_tx.v:23
       :   -- module gen_en_dff declared here: ./utils/gen_dff.v:125
       :   -- module gen_pipe_dff declared here: ./utils/gen_dff.v:18
       :   -- module gen_rst_0_dff declared here: ./utils/gen_dff.v:46
       :   -- module gen_rst_1_dff declared here: ./utils/gen_dff.v:72
       :   -- module gen_rst_def_dff declared here: ./utils/gen_dff.v:98
       :   -- module gen_ticks_sync declared here: ./utils/gen_buf.v:18
       :   -- module i2c_lm75 declared here: ./perips/i2c_lm75.v:14
       :   -- module id declared here: ./core/id.v:21
       :   -- module id_ex declared here: ./core/id_ex.v:20
       :   -- module if_id declared here: ./core/if_id.v:20
       :   -- module jtag_dm declared here: ./debug/jtag_dm.v:27
       :   -- module jtag_driver declared here: ./debug/jtag_driver.v:23
       :   -- module jtag_top declared here: ./debug/jtag_top.v:20
       :   -- module pc_reg declared here: ./core/pc_reg.v:20
       :   -- module pwm declared here: ./perips/pwm.v:16
       :   -- module regs declared here: ./core/regs.v:20
       :   -- module rib declared here: ./core/rib.v:16
       :   -- module tinyriscv declared here: ./core/tinyriscv.v:20
       :   -- module tinyriscv_soc_top declared here: ./soc/tinyriscv_soc_top.v:13
       :   -- module uart_debug declared here: ./perips/uart_debug.v:31
       :   -- module uart_shared declared here: ./perips/uart_shared.v:12
./core/regs.v:74: warning: @* is sensitive to all 32 words in array 'regs'.
```

```
./core/regs.v:86: warning: @* is sensitive to all 32 words in array 'regs'.
./core/regs.v:95: warning: @* is sensitive to all 32 words in array 'regs'.
./perips/pwm.v:75: warning: @* is sensitive to all 4 words in array 'period'.
./perips/pwm.v:76: warning: @* is sensitive to all 4 words in array 'period'.
./perips/pwm.v:77: warning: @* is sensitive to all 4 words in array 'period'.
./perips/pwm.v:78: warning: @* is sensitive to all 4 words in array 'period'.
./perips/pwm.v:79: warning: @* is sensitive to all 4 words in array 'high_time'.
./perips/pwm.v:80: warning: @* is sensitive to all 4 words in array 'high_time'.
./perips/pwm.v:81: warning: @* is sensitive to all 4 words in array 'high_time'.
./perips/pwm.v:82: warning: @* is sensitive to all 4 words in array 'high_time'.
PASS: tb_soc_mem_ops
sim\tb_soc_mem_ops.v:135: $finish called at 12050000 (1ps)
===== EXIT_CODE: 0 =====
```

10_tb_chip_sel - chip_sel 选择逻辑

```
===== COMMAND =====
iverilog -g2012 -Wall -I core -o sim_chip_sel.vvp -f filelist_project.f sim\tb_chip_sel.v fpga\fpga_mem_bridge.v && vvp sim_chip_sel.vvp
===== OUTPUT =====
fpga\fpga_mem_bridge.v:6: warning: timescale for fpga_mem_bridge inherited from another file.
sim\tb_chip_sel.v:1: ...: The inherited timescale is here.
warning: Some design elements have no explicit time unit and/or
       : time precision. This may cause confusing timing results.
       : Affected design elements are:
       :   -- module chip_mem_bridge declared here: ./perips/chip_mem_bridge.v:24
       :   -- module clint declared here: ./core/clint.v:22
       :   -- module csr_reg declared here: ./core/csr_reg.v:20
       :   -- module ctrl declared here: ./core/ctrl.v:21
       :   -- module custom_unit declared here: ./core/custom_unit.v:12
       :   -- module div declared here: ./core/div.v:22
       :   -- module ex declared here: ./core/ex.v:21
       :   -- module full_handshake_rx declared here: ./utils/full_handshake_rx.v:23
       :   -- module full_handshake_tx declared here: ./utils/full_handshake_tx.v:23
       :   -- module gen_en_dff declared here: ./utils/gen_dff.v:125
       :   -- module gen_pipe_dff declared here: ./utils/gen_dff.v:18
       :   -- module gen_rst_0_dff declared here: ./utils/gen_dff.v:46
       :   -- module gen_rst_1_dff declared here: ./utils/gen_dff.v:72
       :   -- module gen_rst_def_dff declared here: ./utils/gen_dff.v:98
       :   -- module gen_ticks_sync declared here: ./utils/gen_buf.v:18
       :   -- module i2c_lm75 declared here: ./perips/i2c_lm75.v:14
       :   -- module id declared here: ./core/id.v:21
       :   -- module id_ex declared here: ./core/id_ex.v:20
       :   -- module if_id declared here: ./core/if_id.v:20
       :   -- module jtag_dm declared here: ./debug/jtag_dm.v:27
       :   -- module jtag_driver declared here: ./debug/jtag_driver.v:23
       :   -- module jtag_top declared here: ./debug/jtag_top.v:20
       :   -- module pc_reg declared here: ./core/pc_reg.v:20
       :   -- module pwm declared here: ./perips/pwm.v:16
       :   -- module regs declared here: ./core/regs.v:20
       :   -- module rib declared here: ./core/rib.v:16
       :   -- module tinyriscv declared here: ./core/tinyriscv.v:20
       :   -- module tinyriscv_soc_top declared here: ./soc/tinyriscv_soc_top.v:13
       :   -- module uart_debug declared here: ./perips/uart_debug.v:31
       :   -- module uart_shared declared here: ./perips/uart_shared.v:12
./core/regs.v:74: warning: @* is sensitive to all 32 words in array 'regs'.
./core/regs.v:86: warning: @* is sensitive to all 32 words in array 'regs'.
./core/regs.v:95: warning: @* is sensitive to all 32 words in array 'regs'.
./perips/pwm.v:75: warning: @* is sensitive to all 4 words in array 'period'.
./perips/pwm.v:76: warning: @* is sensitive to all 4 words in array 'period'.
./perips/pwm.v:77: warning: @* is sensitive to all 4 words in array 'period'.
./perips/pwm.v:78: warning: @* is sensitive to all 4 words in array 'period'.
./perips/pwm.v:79: warning: @* is sensitive to all 4 words in array 'high_time'.
./perips/pwm.v:80: warning: @* is sensitive to all 4 words in array 'high_time'.
./perips/pwm.v:81: warning: @* is sensitive to all 4 words in array 'high_time'.
./perips/pwm.v:82: warning: @* is sensitive to all 4 words in array 'high_time'.
PASS: tb_chip_sel
sim\tb_chip_sel.v:73: $finish called at 2950000 (1ps)
===== EXIT_CODE: 0 =====
```

11_static_disallowed_constructs - 禁用/敏感语法扫描

```
===== COMMAND =====
rg -n \b(casex|casez|force|release|deassign|forever|wait)\b|\$display|\$finish|\$stop|\$readmemh|1'bx|1'bz|32'hx|\binitial\b core debug perips
soc utils fpga
===== OUTPUT =====
fpga\fpga_mem_bridge.v:39:    initial begin
fpga\fpga_mem_bridge.v:43:            $readmemh(ROM_INIT_FILE, rom);
perips\i2c_lm75.v:67:    assign i2c_sda = sda_oe_low ? 1'b0 : 1'bz;
perips\i2c_lm75.v:153:                      sda_oe_low <= ~shifter[bit_cnt]; // release for 1, pull low for 0
perips\i2c_lm75.v:172:                      sda_oe_low <= 1'b0;              // release for ACK
perips\i2c_lm75.v:206:                      sda_oe_low <= 1'b0;              // NACK = release SDA
core\rib.v:9: *   WAIT  : wait for bridge busy to rise and then fall
===== EXIT_CODE: 0 =====
```

12_static_timing_directives - timescale/default_nettype 扫描

```
===== COMMAND =====
rg -n `timescale|timeunit|timeprecision|default_nettype core debug perips soc utils fpga sim
===== OUTPUT =====
sim\tb_chip_sel.v:1:`timescale 1ns/1ps
sim\tb_custom_unit.v:1:`timescale 1ns/1ps
sim\tb_i2c_lm75.v:1:`timescale 1ns/1ps
sim\tb_mem_bridge.v:1:`timescale 1ns/1ps
sim\tb_pwm.v:1:`timescale 1ns/1ps
sim\tb_soc_load.v:1:`timescale 1ns/1ps
sim\tb_soc_mem_ops.v:1:`timescale 1ns/1ps
===== EXIT_CODE: 0 =====
```

13_static_todos_placeholders - TODO/占位项扫描

```
===== COMMAND =====
rg -n TODO|FIXME|XXX|Please replace|replace DEFAULT|signoff|流片|占位|placeholder core debug perips soc utils fpga
===== OUTPUT =====
core\custom_unit.v:8: * Please replace DEFAULT_ID_* bytes with your own student ID before signoff.
===== EXIT_CODE: 0 =====
```

14_static_tristate_inout - inout/三态扫描

```
===== COMMAND =====
rg -n \binout\b|1'bz|\btri\b|\btri1\b|\bwand\b|\bwor\b core debug perips soc utils fpga sim
===== OUTPUT =====
soc\tinyriscv_soc_top.v:38:    inout  wire      i2c_sda
perips\i2c_lm75.v:30:    inout  wire       i2c_sda
perips\i2c_lm75.v:67:    assign i2c_sda = sda_oe_low ? 1'b0 : 1'bz;
sim\tb_i2c_lm75.v:16:    tri1 sda;
sim\tb_chip_sel.v:17:    tri1 i2c_sda;
sim\tb_soc_load.v:22:    tri1 i2c_sda;
sim\tb_soc_mem_ops.v:22:    tri1 i2c_sda;
===== EXIT_CODE: 0 =====
```

15_line_counts - 源码行数统计

```
===== COMMAND =====
internal Python line count
===== OUTPUT =====
core\clint.v	242
core\csr_reg.v	215
core\ctrl.v	71
core\custom_unit.v	192
core\defines.v	173
core\div.v	213
core\ex.v	972
core\id.v	342
core\id_ex.v	111
core\if_id.v	52
core\pc_reg.v	51
core\regs.v	99
core\rib.v	283
core\tinyriscv.v	522
debug\jtag_dm.v	343
debug\jtag_driver.v	295
debug\jtag_top.v	108
perips\chip_mem_bridge.v	111
perips\i2c_lm75.v	252
perips\pwm.v	93
perips\uart.v	329
perips\uart_debug.v	155
perips\uart_shared.v	276
soc\tinyriscv_soc_top.v	306
utils\full_handshake_rx.v	128
utils\full_handshake_tx.v	147
utils\gen_buf.v	46
utils\gen_dff.v	149
fpga\fpga_mem_bridge.v	99
sim\tb_chip_sel.v	75
sim\tb_custom_unit.v	121
sim\tb_i2c_lm75.v	60
sim\tb_mem_bridge.v	86
sim\tb_pwm.v	70
sim\tb_soc_load.v	107
sim\tb_soc_mem_ops.v	137
===== EXIT_CODE: 0 =====
```

16_module_balance - module/endmodule 配平

```
===== COMMAND =====
internal Python module/endmodule balance
===== OUTPUT =====
core\clint.v    module=1        endmodule=1
core\csr_reg.v  module=1        endmodule=1
core\ctrl.v     module=1        endmodule=1
core\custom_unit.v      module=1        endmodule=1
core\defines.v  module=0        endmodule=0
core\div.v      module=1        endmodule=1
core\ex.v       module=1        endmodule=1
core\id.v       module=1        endmodule=1
core\id_ex.v    module=1        endmodule=1
core\if_id.v    module=1        endmodule=1
core\pc_reg.v   module=1        endmodule=1
core\regs.v     module=1        endmodule=1
core\rib.v      module=1        endmodule=1
core\tinyriscv.v        module=1        endmodule=1
debug\jtag_dm.v module=1        endmodule=1
debug\jtag_driver.v     module=1        endmodule=1
debug\jtag_top.v        module=1        endmodule=1
perips\chip_mem_bridge.v        module=1        endmodule=1
perips\i2c_lm75.v       module=1        endmodule=1
perips\pwm.v    module=1        endmodule=1
perips\uart.v   module=1        endmodule=1
perips\uart_debug.v     module=1        endmodule=1
perips\uart_shared.v    module=1        endmodule=1
soc\tinyriscv_soc_top.v module=1        endmodule=1
utils\full_handshake_rx.v       module=1        endmodule=1
utils\full_handshake_tx.v       module=1        endmodule=1
utils\gen_buf.v module=1        endmodule=1
utils\gen_dff.v module=5        endmodule=5
fpga\fpga_mem_bridge.v  module=1        endmodule=1
===== EXIT_CODE: 0 =====
```

17_leftover_vvp - 临时仿真产物清理检查

```
===== COMMAND =====
internal Python leftover .vvp scan
===== OUTPUT =====

===== EXIT_CODE: 0 =====
```

8. 流片前必须补齐的交付物

1. 用课程或 MPW 平台指定工艺库完成综合，提交面积、频率、功耗、约束和 warning/error 报告。
2. 完成 STA，多 corner、多 mode，明确 false path、multicycle path、JTAG_TCK CDC 约束。
3. 完成 CDC/RDC 检查，重点检查 JTAG_TCK 与 core clk、异步 reset 释放、UART RX 异步输入。
4. 实现 pad-level IO Ring wrapper：多 die 共享输出必须由 chip_sel 控制 mux/OE，未选芯片输出必须高阻或不驱动。
5. 补 DFT/scan/测试模式，按课程或平台要求插入 scan chain。
6. 完成 APR、天线、IR/EM、DRC、LVS，提供 clean GDS。
7. 替换 core/custom_unit.v 的 DEFAULT_ID_* 学号占位。
8. 扩展验证：RV32I/M 指令回归、CSR/exception/mret、中断、JTAG transaction、UART downloader 与助教脚本、真实 FPGA 板级。